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# COMPENDIUM OF INDIAN STANDARDS ON IS 101 SERIES

## (METHODS OF SAMPLING AND TEST FOR PAINTS, VARNISHES AND RELATED PRODUCTS)


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# INTRODUCTION

The Indian Standard IS 101 is a comprehensive series developed by the Bureau of Indian Standards (BIS) to establish standardized methods for sampling and testing paints, varnishes, and related products. This series is crucial for ensuring the quality, performance, and safety of coatings used in various applications, including construction, automotive, and industrial Sectors.

The IS 101 series provides standardized procedures that help manufacturers, quality control laboratories, and regulatory bodies ensure that paints and coatings meet specified performance criteria. This standardization is essential for maintaining product quality, ensuring safety, and facilitating trade both domestically and internationally.

IS 101 "Methods of test for ready mixed paints and enamel", initially, was published as a unified standard in 1950. The standard was subsequently revised in 1961, 1964 and 1987. During the third revision, recognizing the need for clarity, the committee decided to restructure it by splitting into various parts based on type of tests. These parts included tests on liquid paints (general and physical), chemical examination, film formation, optical assessments, and mechanical tests on paint film formation. Each part was further subdivided into Sections, addressing specific tests within those categories.

## OVERVIEW OF IS 101 SERIES

The IS 101 series is divided into multiple parts and Sections, each focusing on specific aspects of testing and evaluation:

## PART 1: TESTS ON LIQUID PAINTS (GENERAL AND PHYSICAL)

- **Section 1: Sampling**

  This standard prescribes methods of sampling for paints, varnishes and related products. It is one of a series of standard dealing with sampling and testing of these products. This standard aims at obtaining uniform samples of convenient size and adequately representative of the product being sampled. The sample or samples so obtained are suitable for examination and preparation prior to testing.

- **Section 2: Preliminary examination and preparation of samples for testing/ISO 1513:2010**

  This standard specifies both the procedure for preliminary examination of a single sample, as received for testing, and the procedure for preparing a test sample by blending and reduction of a series of samples representative of a consignment or bulk of paint, varnish or related product.

- **Section 3: Preparation of Panels**

  This standard prescribes procedures for the preparation, prior to painting of standard panels for use in general methods of test for paints, vanishes and related products. This standard covers the following types of standard panels.

  a) Steel panels;
  b) Tin plate panels;
  c) Aluminium panels;
  d) Wood panels; and
  e) Cement/Asbestos panels.

- **Section 4: Brushing Test**

  This standard prescribes procedures for assessing the brushing and flow characteristics of paints and related materials when applied to relatively large areas of closely defined substrates. It can also be used to observe other properties like the tendency of the paint to retract from sharp edges and protrusions with consequent loss of opacity and protective power.

- **Section 5: Consistency**

  This standard prescribes three methods for determining the dynamic viscosity of paints and related products. The methods are:
  a) Flow cup method;
  b) Cone and plate or concentric cylinder viscometer method; and
  c) Stromer viscometer method.

Consistency of a paint material is judged by inserting a clean metal rod or palette knife into the original container and shall be in such a condition that stirring easily produces a smooth uniform paint suitable for application by specified method.

- **Section 6: Flash Point**

  This standard prescribes a test and determination of flash point by closed cup equilibrium method. This standard prescribes flash/no flash test and actual determination, and is applicable between the temperature range of 5 °C and 110 °C.

  In case of flash/no flash test, the sample is heated in a suitably designated closed cup in a suitable water bath. The ignition trial is carried out after the test portion has been maintained under equilibrium conditions for at least 10 minutes at the selected equilibrium temperature.

- **Section 7: Mass per 10 litres - determination of density - pycnometer method/ISO 2811-1:2016**

  This standard specifies a method for determining the density of paints, varnishes and related products using a metal or Gay-Lussac pycnometer. The method is limited to materials of low or medium viscosity at the temperature of test. The Hubbard pycnometer (see ISO 3507) can be used for highly viscous materials.

- **Section 8: Pigments and extenders determination of pH value of an aqueous suspension/ISO 787-9:2019**

  This document specifies a general method of test for determining the pH value of an aqueous suspension of a sample of pigment or extender.

## PART 2: TESTS ON LIQUID PAINTS (CHEMICAL EXAMINATION)

- **Section 1: Water content**

  This standard prescribes the Dean and Stark method and the Karl Fischer (Direct Electrometric Titration) method for determination of water content in paints.

  **Karl Fischer Method** — Reaction of any water present in a test portion with a solution of Karl Fischer reagent, previously standardized by titration with an exactly known mass of water.

  **Dean and Stark Method** — The material is refluxed with organic solvent which is immiscible with water. The carrier liquid distils into a graduated receiver carrying water with it which then separates to form the lower layer, the excess carrier liquid overflowing from the trap and returning to the still.

- **Section 2: Volatile matter**

  The standard prescribes the method for the determination of volatile and non-volatile matter in paint, varnishes and related product.

- **Section 3: Determination of Volatile Organic Compound (VOC) Content – Difference Method/ISO 11890-1:2020**

**Principle**: This method assumes that the volatile matter is either water or organic. After preparation of the sample, the non-volatile matter content is determined in accordance with ISO 3251, and the water content is determined using a titration technique employing Karl Fischer reagent in accordance with ISO 760. The contents of exempt compounds, if applicable, are then determined using the method specified in ISO 11890-2. The VOC content of the sample is then calculated.

- **Section 4: Determination of volatile organic compound (VOC) and or semi volatile organic compounds (SVOC) content gas-chromatographic method/ISO 11890-2:2020**

  **Principle:** After preparation of the sample, VOCs, SVOCs and NVOCs are separated by the gas chromatographic technique. Either a hot or a cold sample injection system is used, depending on the sample type. Hot injection is the preferred method. After the compounds have been identified, e.g. via GC-MS, they are quantified from the peak areas with respect to their compound specific relative response with the help of an internal standard, via GC-FID. Non-identifiable substances are quantified with respect to a surrogate standard that can be identical to the internal standard. A calculation is performed to give the VOC and/or SVOC content of the sample.

- **Section 5: Determination of volatile organic compound content of low-VOC emulsion paints (In-Can VOC)/ISO 17895:2005**

  This standard specifies a gas-chromatographic method of quantitatively determining the volatile organic compound content of low content emulsion paints (in-can VOC).

  **Principle**: The VOCs in a very small amount of thinned sample are fully evaporated in a headspace injector and then determined by gas-chromatographic analysis, as follows: A few microlitres of the sample diluted with buffer solution are heated to 150 °C in a septum-sealed vial and, when fully evaporated, some of the vapour phase is transferred to a non-polar capillary column. The peak areas of all the components with retention times less than that of tetradecane (boiling point 252.6 °C) are integrated. Standard additions of a stock reference compound mixture are employed at four concentration levels to determine the VOC content. The result is based on the average response factor of the reference compound mixture.

## PART 3 TEST ON PAINT FILM FORMATION

- **Section 1: Drying time**

  This standard prescribes the method for determination of surface dry, hard dry and tack-free dry of paints, varnishes and related products.

- **Section 2: Film thickness**

  This standard prescribes methods for measuring the film thickness of paints and related materials. Method used to determine dry film thickness are as follows;

  a) Micrometer Method;
  b) Dial Guage Method;
  c) Microscope Method; and

d) Non Destructive Instrumental Method.

- **Section 4: Finish**

  This standard prescribes the details for the determination of finish if paints and varnish films

- **Section 5: Determination of fineness of grind/ISO 1524:2020**

  This document specifies a method for determining the fineness of grind of paints, inks and related products by use of a suitable gauge, graduated in micrometres.

  It is applicable to all types of liquid paints and related products, except products containing pigments in flake form (e.g. glass flakes, micaceous iron oxides, zinc flakes).

- **Section 6: Determination of the Surface Tension of Liquid Using the Pendant Drop Method/ISO 19403-3:2017**

  This standard specifies a test method to measure the surface tension of liquids with an optical method using the pendant drop. One drop of the respective liquids to be tested is captured hanging from a needle, where the drop shall deviate significantly from the spherical shape due to its own mass. The surface tension is calculated from the shape of the pendant drop in accordance with the Young-Laplace equation. The polar and disperse fractions of the surface tension can be determined with at least two methods, which are specified in ISO 19403-4 and ISO 19305-5.

## PART 4: OPTICAL TESTS

- **Section 1: Opacity**

  This standard prescribes three methods for determination of contracts ratio (opacity)

  1) **Method using Black and White Charts** — This method is based on the observation that contrast ratio is an approximately linear function of the reciprocal film thickness. Over a restricted film thickness range which also corresponds to that used for normal application of white or light coloured paints. It is thus possible to interpolate graphically or by computation between results on films of different thicknesses, with satisfactory accuracy.
  2) **Method using Polyester film** — This method is also very similar to the one for determination of opacity by the method using black and white charts. This method prescribes the determination of opacity of given paint films applied at a spreading capacity of 20 $m^2/l$ to colourless transparent polyester film, the reflectance being measured subsequently over agreed black and white glass plates.
  3) **Pfund Cryptometer Method** — The cryptometer is an instrument designed to measure the wet opacity of the paint. It consists of a calibrated wedge and enables the operator to determine the thickness of the paint required for complete obliteration of a contrasting black and white surface. This can be converted to wet opacity per litre.

- **Section 2: Colour-visual comparison of colour of paints**

This document specifies a method for the visual comparison of the colour of films of paints or related products against a standard (either a reference standard or a freshly prepared standard) using artificial light sources in a standard booth.

**Principle**: The colours of the paint films to be compared are observed under specified illumination and viewing conditions, using a colour matching booth. For the expression of colour difference components (hue, chroma and lightness) a procedure is described, i.e. description by using a particular rating scheme. Assessment of metamerism is also taken into account

- **Section 3: Light fastness test**

This standard prescribes a method for the determination of light fastness of paints for interior use by exposure to light from artificial sources under prescribed conditions. In case of dispute, the xenon arc shall be used as referee method.

- **Section 4: Gloss — Determination of gloss value at 20°, 60° and 85°/ISO 2813: 2014**

This Standard specifies a method for determining the gloss of coatings using the three geometries of 20°, 60° or 85°. The method is suitable for the gloss measurement of non-textured coatings on plane, opaque substrates.

With a reflectometric apparatus, gloss values are determined on coated surfaces, correlating with the visual gloss perception. In this context (glossmeter), the ratio of the gloss of the coating and the gloss of a polished plane glass plate with specified reference refractive index is obtained.

## PART 5: MECHANICAL TESTS ON PAINT FILMS

- **Section 1: Hardness Tests**

This standard prescribes the following hardness tests for films formed by paints and allied products

1) **Pendulum Test** — The resistance of a single coat film or multicoat system of paints, varnishes and related products is determined to a single blow impact by a pendulum. The method of test is to be completed by giving supplementary information as follows: Material and surface preparation of the substrate; Method of application; The thickness and coating; and Duration and conditions of coated panel.
2) **Indentation Test** — A method known as Buchholz indentation test is carried out on single coat or multicoat systems. The length of indentation produced when an indentor of specified size and shape is applied to the coating under defined conditions is indicative of the residual deformation of the coating. The result is expressed as the reciprocal of the indentation length and this increases as the desired property increases.
3) **Pencil Hardness Test** — This method is primarily intended for use in the production shop and as such test panels are not normally used.
4) **Pressure Test** — In this test, two painted film after specified drying period, facing each other and in close contact, are tested under pressure.

- **<u>Section 2: Flexibility and Adhesion</u>**

Prescribes the following four tests for determination of flexibility and adhesion:

a) **Bend test** — This is an empirical test for assessing the resistance of a coating of paint, varnish, etc, to cracking and/or detachment from a metal substrate when subjected to bending round a cylindrical mandrel.

b) **Scratch hardness test** — This is a test method for determining the resistance of a paint film to penetration by scratching with a needle. The test may be used as a GO/NO-GO test by applying a single specified load to the needle. It may also be used by applying increasing loads to the needle to determine the maximum load at which the coating is pulled/penetrated to the substrate.

c) **Cupping test** — This is a test for assessing the paint film to cracking and/or detechment from a metal substrate when subjected to gradual deformation by indentation under specified conditions. The method may be either used as a GO/NO-GO test by testing to a specified depth of indentation to assess compliance with a particular requirement or by gradually increasing the depth of indentation to determine the minimum depth at which the film cracks and/or becomes detached from the substrate.

d) **Pull off test** — This is a pull off test on a single or multi coat system of paint, varnish or related products, one test may be carried out using different substrates.

e) **Cross Cut Adhesion Test** — This method is applicable for the assessment of the adhesion of the coating films to metallic substrates by applying and removing the 25 mm semi-transparent pressure sensitive tape over cuts made in the film.

- **<u>Section 3: Impact resistance - falling - weight test, large - Area indenter/ISO 6272-1: 2011</u>**

This standard prescribes a method for evaluating the resistance of a dry film of paint, varnish or related product to cracking or peeling from a substrate when it is subjected to a deformation caused by a falling weight, with a 20-mm-diameter spherical indenter, dropped under standard conditions.

The coating under test is applied to suitable, thin (normally metal) panels. After the coating has cured, a standard weight is dropped on to each panel from a height that will cause deformation of the coating and the substrate. The test can be carried out with the coated side of the panel facing upwards (i.e. towards the falling weight) or downwards (i.e. away from the weight). By gradually increasing the height from which the weight drops, the point at which failure occurs can be determined. Films generally fail by cracking, which is made more visible by the use of a magnifier.

- **<u>Section 4: Print-Free Test</u>**

This standard prescribes a method for assessing, by means of a simple empirical test, the resistance of coat of paint, varnish or related products to imprintability by a nylon gauze under a specified fort applied for a specified time.

- **<u>Section 5: Impact resistance - Falling - weight test, small - area indenter/ISO 6272-2:2011</u>**

  This standard prescribes a method for evaluating the resistance of a dry film of paint, varnish or related product to cracking or peeling from a substrate when it is subjected to a deformation caused by a falling weight, dropped under standard conditions, acting on a small-area spherical indenter.

  **Principle**: The coating under test is applied to suitable thin metal panels. After the coatings have cured, a standard weight is dropped a distance so as to strike an indenter that deforms the coating and the substrate. The test can be carried out with the coated side of the panel facing upwards (i.e. towards the falling weight and indentor) or downwards (i.e. away from the weight and indentor). By gradually increasing the distance the weight drops, the point at which failure occurs can be determined. Films generally fail by cracking, which is made more visible by the use of a magnifier or, on steel, by the application of a copper sulfate solution.

- **<u>Section 6: Determination of resistance to abrasion by abrasive-paper covered wheels and rotating test specimen/ISO 7784-1:2023</u>**

  This standard specifies a method for determining the resistance to abrasion of coatings, for which two loaded, freely rotatable but eccentricity arranged abrasive-paper covered wheels affect the coating of the rotating test specimen.

  **Principle**: Two rubber wheels, covered with agreed abrasive paper, are mounted to pivoting arms and pressed onto the coating of the rotating test specimen applying the agreed test load. The eccentric arrangement of the axes of the abrasive wheels relative to the axis of rotation causes a crosswise abrasive wear in a ring-shaped zone. The loss of mass of the coating caused by abrasive wear after the agreed number of cycles is determined.

- **<u>Section 7: Determination of resistance to abrasion by abrasive rubber wheels and rotating test specimen/ISO 7784-2:2023</u>**

  This standard specifies a method for determining the resistance to abrasion of coatings, for which two loaded, freely rotatable but eccentricity arranged abrasive rubber wheels affect the coating of the rotating test specimen.

  **Principle**: Two abrasive rubber wheels are mounted to pivoting arms and pressed onto the coating of the rotating test specimen applying the agreed test load. The eccentric arrangement of the axes of the abrasive wheels relative to the axis of rotation causes a crosswise abrasive wear in a ring-shaped zone. The loss of mass of the coating caused by abrasive wear after the agreed number of cycles is determined.

- **<u>Section 8: Determination of resistance to abrasion by abrasive-paper covered wheel and linearly reciprocating test specimen/ISO 7784-3:2023</u>**

  This standard specifies a method for determining the resistance to abrasion of coatings, for which a loaded, rigid abrasive-paper covered wheel affects the coating of the linearly reciprocating test specimen.

**Principle**: A rigid abrasive wheel, covered with abrasive paper, is pressed onto the coating applying the test load. The test specimen is reciprocated with specified stroke length and double-stroke frequency. The abrasive wheel itself is rotated by a small angle after each double stroke, so that a new fresh portion of the abrasive paper is applied. The specimen is set with its testing surface facing downward, and the testing surface is abraded from underneath.

## PART 6: DURABILITY TESTS ON PAINT FILMS

- **Section 1: Resistance to Humidity under Conditions of Condensation**

This standard prescribes methods for determination of resistance to humidity under continuous condensation. Also covers methods on neutral and artificial salt spray tests. This test is carried by suspending the painted panel after specified period of drying in a corrosion cabinet maintained at 100 percent relative humidity and a temperature cycle of 42 °C to 48 °C for seven days and examining it for any signs of deterioration and corrosion of metal surface.

- **Section 2: Keeping Properties**

This standard prescribes the method to determine the change in certain properties that may take place when packaged paint of either the solvent thinned or latex type is stored at normal room conditions.

- **Section 3: Moisture vapour permeability**

This standard prescribes the method to determine the rate at which moisture passes through films of paint, varnish and related products.

- **Section 4: Degradation of coatings (pictorial aids for evaluation)**

This standard prescribes the method for evaluation of the degree of flaking ( scaling ) and blistering, that may develop in the paint system, by comparison with photographic standards.

- **Section 5: Accelerated weathering test**

This standard prescribes the following three methods for accelerated weathering tests:
a) Carbon arc,
b) Xenon arc, and
c) UV condensation type.

## PART 7: ENVIRONMENTAL TESTS ON PAINTS FILMS

- **Section 1: Resistance to water**

This standard prescribes a method for determination of the resistance to water of a paint. This method gives an indication of the results likely to be obtained when painted articles are stored under conditions where prolonged condensation may be produced but not an extremely corrosive atmosphere.

- **Section 2: Resistance to liquids**

This standard prescribes general methods for determining the resistance of single-coat paint films, paint systems or allied products to the action of liquids.

Three methods of test are specified and the method to be used depends on the particular requirements of the test material. Method I (Immersion method) is intended for more resistant coatings, requiring longer periods of exposure than those which may be tested by Method 2 (Using an absorbent medium) or Method 3 (Spotting method).

- **Section 3: Determination of the effect of heat/ISO 3240:2016**

This Standard specifies a method for determining the resistance of single coatings or multi-coat systems of paints, varnishes or related products to changes in gloss and/or colour, blistering, cracking and/or detachment from the substrate under conditions of a specified temperature. This procedure is applicable to products intended for use on domestic radiators or other articles likely to be subjected to similar temperatures.

- **Section 4: Resistance to bleeding of pigments**

This standard prescribes a method for comparing the resistance to bleeding of a pigment with that of an agreed sample.

## PART 8: TESTS FOR PIGMENTS AND OTHER SOLIDS

- **Section 1: Residue on sieve**

This standard prescribes the method of determination of residue on sieves for paints, varnishes and related products.

- **Section 2: Pigments, non-volatile vehicle and non-volatile matter**

This standard prescribes the methods of sampling and test for determination of pigment and non-volatile matter content in paints, varnishes and other related products.

- **Section 3: Ash content**

This standard prescribes the method to determine the ash content of the packaged paints, varnish and related products.

- **Section 4: Phthalic anhydride**

This standard prescribes the method to determine the phthalic anhydride content in the paint.

- **Section 5: Lead restriction test**

This standard prescribes methods of test for lead restriction in paints and allied products. For lead restriction test any one of the following five methods may be used:

a) **Electrolysis method**;

b) **Molybdate method** — Paint is digested with concentrated sulphuric acid and nitric acid in order to convert lead to lead sulphate followed by extraction with ammonium acetate. Finally, lead is precipitated as lead molybdate and weighed as lead molybdate.

c) **Sulphide method** — Determination of lead in lead restricted paints is carried cut by precipitating the lead as sulphide from the separated pigment, which is finally oxidized to lead sulphate.

d) **Atomic Absorption Spectroscopy method (AAS)** — This method covers the determination of lead content in pigmented coating, it is also applicable for varnish and lacquer. This method is not applicable for determination of lead in sample containing antimony pigment as low recovery is obtained.

e) **Inductively Coupled Plasma-Optical Emission Spectrometry (ICP-OES)** — Lead in wet paint (in a can) or dried paint sample (film, chips, powder, etc.) is acid extracted (solubilized) by digestion with nitric acid and hydrogen peroxide aided by heating or nitric acid facilitated by microwave energy. The lead content of the digested sample is then analysed using ICP-OES.

- **Section 6: Volume solids**

This standard prescribes the method to determine volume solids in paints and allied products. This method is intended to provide a measure of the volume of dry coating obtainable from a given volume of liquid coating. This volume -is considered to be the most equitable means of comparing the coverage and the wet film thickness of the given paint.

## PART 9: TESTS FOR LACQUERS AND VARNISH

- **Section 1: Acid value**

This standard prescribes the method to determine the acid value of the packaged paint, varnish and related products. Known mass of the material is dissolved in a neutral solvent mixture of toluene and ethanol acid titrated against standard potassium hydroxide solution using phenolphthalein as indicator.

- **Section 2: Rosin test**

This standard prescribes the method to determine the freedom from rosin in the packaged paint, varnish and related products.
The material shall be tested for freedom from rosin by either of the following two methods:
a) **Liebermann-Starch test** — This test method covers procedures for the qualitative detection of rosin in varnishes and extracted vehicle from paints by the Liebermann Starch Test. Rosin may be present as either free rosin (abietic acid), esterified rosin or as metal salt.
b) **Halphen-Hicks test** — Acetylated and solvent extracted material with rosin as impurity when treated with phenol, develops specific colour with bromine vapours. This reaction is used as basis of detection of rosin.

## PART 10: INSTRUMENTAL ANALYSIS

- **Section 1: Particle size analysis – laser diffraction methods/ISO 13320:2020**

  The laser diffraction or scattering technique for the determination of particles size distributions, PSDs, is based upon the phenomenon that the angular distribution of the intensity of scattered light by a particle (scattering pattern) is dependent on the particle size. When the scattering is from a cloud or ensemble of particles the intensity of scattering for any given size class is related to the number of particles and their optical properties, present in that size class.

- **Section 2: Electrochemical impedance spectroscopy (eis) on coated and uncoated metallic specimens – terms and definitions/ISO 16773-1:2016**

  This standard defines terms for electrochemical impedance spectroscopy (EIS) for use in the other parts of ISO 16773.

- **Section 3: Electrochemical impedance spectroscopy (EIS) on coated and uncoated metallic specimens - Collection of data/ISO 19773-2:2016**

  This standard gives guideline for optimizing the collection of EIS data with focus on high impedance systems. High impedance in the context of intact coatings refers to systems with an impedance greater than $10^9$ Ω·cm2. This does not preclude measurements on systems with lower impedance. For uncoated samples extra information can be found in ISO/TR 16208.

- **Section 4: Electrochemical impedance spectroscopy (EIS) on coated and uncoated metallic specimens - Processing and analysis of data from dummy cells/ISO 19773-3:2016**

  This standard specifies a procedure for the evaluation of the experimental set-up used for carrying out EIS on high-impedance coated samples. For this purpose, dummy cells are used to simulate high-impedance coated samples. On the basis of the equivalent circuits described, this part of ISO 16773 gives guidelines for the use of dummy cells to increase confidence in the test protocol, including making measurements, curve fitting and data presentation.

- **Section 5: Electrochemical impedance spectroscopy (EIS) on coated and uncoated metallic specimens - Examples of spectra of polymer - Coated and uncoated specimens/ISO 19773-4:2016**

  This standard gives some typical examples of impedance spectra of polymer-coated and uncoated specimens. Some guidance on interpretation of such spectra is also given. Further examples of spectra of low-impedance systems (range from, e.g. 10 Ω to 1 000 Ω) are given in ISO/TR 16208 and in ASTM G106. ISO 1677-2 gives guidelines for optimizing the collection of EIS data with focus on high-impedance systems.

## PART 11: EVALUATION OF DEGRADATION OF COATINGS DESIGNATION OF QUANTITY AND SIZE OF DEFECTS AND OF INTENSITY OF UNIFORM CHANGES IN APPEARANCE

- **Section 1: General introduction and designation system/ISO 4628-1:2016**

  This standard defines a system for designating the quantity and size of defects and the intensity of changes in appearance of coatings and outlines the general principles of the system used throughout ISO 4628. This system is intended to be used, in particular, for defects caused by ageing and weathering, and for uniform changes, for example yellowing.

- **Section 2: Assessment of degree of blistering/ISO 4628-2:2016**

  This standard specifies a method for assessing the degree of blistering of coatings by comparison with pictorial standards. The pictorial standards provided in this standard illustrate blisters in the sizes 2, 3, 4, and 5, and each size in the quantities (densities) 2, 3, 4, and 5.

- **Section 3: Assessment of degree of rusting/ISO 4628-3:2016**

  This standard specifies a method for assessing the degree of rusting of coatings by comparison with pictorial standards. The pictorial standards provided in this standard show coated steel surfaces which have deteriorated to different degrees by a combination of rust broken through the coating and visible underrust.

- **Section 4: Assessment of degree of cracking/ISO 4628-4:2016**

  This standard specifies a method for assessing the degree of cracking of coatings by comparison with pictorial standards.

- **Section 5: Assessment of degree of flaking/ISO 4628-5:2022**

  This standard specifies a method for assessing the degree of flaking of coatings by comparison with pictorial standards.

- **Section 6: Assessment of degree of chalking by tape method/ISO 4628-6:2023**

  This document provides pictorial reference standards for designating the degree of chalking of paint coatings. It also describes a method by which the degree of chalking is rated. In using this method, it is essential that care be taken to distinguish between true degradation products and adhering dirt, particularly when chalking is slight.

  **Principle**: The chalking is removed from the coating under test using an adhesive tape. The chalking adhering to the tape is examined against a contrasting background (either black or white, whichever gives the greater contrast) and the degree of chalking is assessed with reference to a rating scale.

- **Section 7: Assessment of degree of chalking by velvet method/ISO 4628-7:2016**

  This standard specifies a method suitable, in particular, for rating the degree of chalking on white or coloured exterior coatings and coating systems on rough surfaces (i.e. those having a roughness greater than segment 4 of the reference comparator G as described in ISO 8503-1.

**Principle**: Loosely adherent powder is removed from the coating under test, using a suitable fabric. The degree of chalking is assessed with reference to a rating scale.

- **Section 8: Assessment of degree of delamination and corrosion around a scribe or other artificial defect/ISO 4628-8:2012**

  This standard specifies a method for assessing delamination and corrosion around a scribe or other artificial defect on a coated panel or other coated test specimen, caused by a corrosive environment.

  **Principle:** The delamination around a scribe or other artificial defect is assessed either directly at the end of the test period, immediately after removal of the test panel from a previous conditioning environment, or after conditioning for a specified period. The corrosion around the scribe or other artificial defect is assessed either immediately after removal of the test panel from a previous conditioning environment or after removal of the coating. Both the degree of delamination and the degree of corrosion are determined by measurement and calculation.

- **Section 10: Assessment of degree of filiform corrosion/ISO 4628-10:2024**

  This standard specifies a method for assessing the amount of filiform corrosion developed from a scribed mark by measuring the length of the longest filament L and the most frequent length M of filaments.